A binary $[50, 40]_2$ code of covering radius 2, and a new seed for the $R = 2$ iterative family

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Abstract

We exhibit a binary linear $[50, 40]_2$ code of covering radius 2, showing $\ell_2(10, 2) \leq 50$ and improving by one the length 51 of the Kaikkonen–Rosendahl code of 2003, which is the entry recorded for $r = 10$ in Table 5.1 of Davydov, Marcugini and Pambianco (arXiv:2511.02542, November 2025). The parity-check matrix admits a $(2, 0)$-partition into $p(\boldsymbol{H}) = 10$ subsets, one fewer than the computer-searched $p(\boldsymbol{H}_{KR}) = 11$ of that paper. Since their infinite $R = 2$ family is seeded by the $r = 10$ entry, substituting this code moves the whole iteration: Construction QM_2^2 gives $[815, 797]_2 2$ and $[1631, 1611]_2 2$ codes at $r = 18$ and $r = 20$, against the published 831 and 1663, and iterating gives $n = 51 \cdot 2^{r/2-5} - 1$ for the reachable even r, with asymptotic covering density $51^2/2^{11} \approx 1.27002$ in place of $26^2/2^9 \approx 1.32031$. All claims are exhaustively machine-checked from the committed matrices; verification code accompanies the note.

1 Introduction

Let $\ell_2(r, R)$ denote the smallest length of a binary linear code with codimension r and covering radius R. If $\boldsymbol{H}$ is an $r \times n$ parity-check matrix with column set $S \subset \mathbb{F}_2^r$, then the code has covering radius at most 2 if and only if

$$\{0\} \cup S \cup (S + S) = \mathbb{F}_2^r. \tag{1}$$

The covering density of an $[n, n-r]_2 2$ code is $\mu = \left(1 + n + \binom{n}{2}\right)/2^r$, and $\bar{\mu}(2) = \liminf_{r \to \infty} \mu$ along a family.

Davydov, Marcugini and Pambianco [1] recently improved the upper bounds on $\ell_2(r, 2)$ for many r and produced a new infinite family with $\bar{\mu}(2) \approx 1.32031$, superseding the value ≈ 1.42383 that had stood since 1992. Their Table 5.1 records, for $r = 10$, the length 51 of the Kaikkonen–Rosendahl code [2]; that entry is not improved in their paper.

The $r = 10$ entry is load-bearing. Their family ([1], Theorem 5.7) has lengths

$$\Phi(r) = 26 \cdot 2^{r/2-4} - 1 = 2^m(51 + 1) - 1, \qquad m = r/2 - 5,$$

which is precisely the Kaikkonen–Rosendahl length 51 pushed forward by Construction QM_2^2; their Theorem 5.7(i) states that the 2-partitions of $\boldsymbol{H}_{KR}$ are "very important for the next iterative process". A shorter seed admitting a small enough 2-partition therefore re-seeds the entire family.

2 The code

Theorem 1. *Let $\boldsymbol{H}$ be the 10×50 binary matrix listed in Appendix A. Then $\boldsymbol{H}$ has $\mathbb{F}_2$-rank 10, its 50 columns are distinct and nonzero, and its column set satisfies (1). Hence $\boldsymbol{H}$ is the parity-check matrix of a binary $[50, 40, 3]_2 2$ code of covering radius exactly 2, and*

$$\ell_2(10, 2) \leq 50.$$

Its covering density is $\mu = 319/256 = 1.24609375$, against $1327/1024 \approx 1.29590$ for the Kaikkonen–Rosendahl code.

Proof. Finite verification. Enumerating the empty sum, the 50 singletons and all $\binom{50}{2} = 1225$ unordered pairs of columns and tabulating the resulting 1276 syndromes shows that all 1024 elements of $\mathbb{F}_2^{10}$ occur, with multiplicity histogram $\{1{:}859,\ 2{:}129,\ 4{:}24,\ 5{:}9,\ 6{:}3\}$. Exactly 973 syndromes are neither 0 nor a column of $\boldsymbol{H}$, so the covering radius is not 1; it is therefore exactly 2. The rank and distinctness claims are immediate from the listing. Minimum distance 3 follows from the 10 linearly dependent triples reported in Proposition 2. $\square$

Proposition 2. *With $\boldsymbol{H}$ as in Theorem 1:*

(i) Deleting any single column of $\boldsymbol{H}$ leaves at least 9 elements of $\mathbb{F}_2^{10}$ uncovered; the best deletions, of the columns 381, 479 and 927, leave exactly 9. Consequently the column set is a minimal 1-saturating set in $PG(9, 2)$, equivalently a locally optimal covering code.

(ii) $\boldsymbol{H}$ has minimum distance $d = 3$ and exactly 10 linearly dependent triples of columns.

Proof. Finite verification; for (i), 50 recounts over all 2^{10} syndromes. $\square$

3 A 2-partition with $p(\boldsymbol{H}) = 10$

We use Definition 3.2 of [1]: a partition of the column set of $\boldsymbol{H}$ into nonempty subsets is a $(2,0)$-partition if every element of $\mathbb{F}_2^r$, including 0, is a sum of at most two columns of $\boldsymbol{H}$ belonging to distinct subsets, and $p(\boldsymbol{H})$ denotes the number of subsets.

Theorem 3. *The partition $\mathscr{P}$ of the columns of $\boldsymbol{H}$ into the ten subsets*

$$\{2,128,202,212,771,855,897,981\},\ \{86\},\ \{381,893,1003\},\ \{183,297\},$$
$$\{1,65,247,256,320,438,502,734\},\ \{15,173,329,366,460,559,653,846\},$$
$$\{4,16,391,491,742,754,869,881\},\ \{479,1004\},\ \{403\},$$
$$\{32,341,373,576,608,777,789,821,927\}$$

(columns written as integers, cf. Appendix A) is a $(2,0)$-partition, so $p(\boldsymbol{H};\mathscr{P}) = 10$. Moreover the columns 491, 734, 821 sum to the zero column and lie in three distinct subsets of $\mathscr{P}$.

Proof. Finite verification. Of the 1024 syndromes, one is zero and 50 are columns of $\boldsymbol{H}$, both admissible by Definition 3.2; each of the remaining 973 is realised by a pair of columns from distinct subsets. For the last claim, $491 \oplus 734 \oplus 821 = 0$ and these lie in the seventh, fifth and tenth subsets as listed. □

By comparison, [1] Theorem 5.2 obtains $p(\boldsymbol{H}_{KR}) = 11$ by computer search. The final assertion of Theorem 3 is the analogue of their Theorem 5.2(ii), which they require in order to apply Construction QM_5^2 at $r = 28$.

Remark 4. The partition is tight in the following sense: 821 of the 973 syndromes requiring a pair have a *unique* such representation, and each of those forces its two columns into distinct subsets.

4 Propagation

Construction QM_2^2 ([1], Theorem 4.1, equations (4.2) and (4.4)) takes an $[n_0, n_0 - r_0]_2 2$ code with a 2-partition into $p(\boldsymbol{H}_0)$ subsets, an indicator set $\mathscr{B} = \mathbb{F}_{2^m}$ subject to $n_0 \geq 2^m \geq p(\boldsymbol{H}_0)$, and $\boldsymbol{D} = \boldsymbol{D}_1(2)$, and returns an $[n, n-r]_2 2$ code with

$$n = 2^m(n_0+1) - 1, \qquad r = r_0 + 2m, \qquad p(\boldsymbol{H}_C) \leq 2^{m+1} + 1.$$

With $n_0 = 50$ and $p(\boldsymbol{H}_0) = 10$ the condition $n_0 \geq 2^m \geq p(\boldsymbol{H}_0)$ holds exactly for $m = 4$ and $m = 5$.

Theorem 5. *Applying* QM_2^2 *to the code of Theorem 1 with the partition of Theorem 3 yields*

$$\begin{aligned} m = 4: &\quad [815, 797]_2 2,\ r = 18,\ \mu = 332521/262144 \approx 1.26847, \\ m = 5: &\quad [1631, 1611]_2 2,\ r = 20,\ \mu = 1330897/1048576 \approx 1.26924, \end{aligned}$$

so $\ell_2(18, 2) \leq 815$ *and* $\ell_2(20, 2) \leq 1631$. *The published values are* $\Phi(18) = 831$ *and* $\Phi(20) = 1663$.

Proof. The two matrices are constructed explicitly and each is verified by complete enumeration of $\mathbb{F}_2^{18}$ and $\mathbb{F}_2^{20}$ respectively: 262144/262144 and 1048576/1048576 syndromes are covered, and both matrices have full rank. □

Corollary 6. *Iterating* QM_2^2 *from the seed of Theorem 1, a state* (r, p) *with length* $n(r) = 51 \cdot 2^{r/2-5} - 1$ *admits a step with parameter* m *whenever* $n(r) \geq 2^m \geq p$, *producing codimension* $r + 2m$ *and* $p \leq 2^{m+1} + 1$. *The even* $r \leq 64$ *so reachable are*

10, 18, 20, 30, 32, 34, 36, 38, 40, 46, 48, 50, 52, 54, 56, 58, 60, 62, 64,

with $\ell_2(r, 2) \leq 51 \cdot 2^{r/2-5} - 1$ *and*

$$\mu(r) = \frac{51^2}{2^{11}} - \frac{51}{2^{r/2+1}} + \frac{1}{2^r} \nearrow \frac{51^2}{2^{11}} = \frac{2601}{2048} \approx 1.27002.$$

Hence $\bar{\mu}(2) \leq 2601/2048$, *and since Green's* $f(2)$ *is a* $\liminf$ *over* r, *also* $f(2) \leq 2601/2048$.

The even $r \in \{12, 14, 16, 22, 24, 26, 28, 42, 44\}$ are *not* reachable by QM_2^2 from this seed. In [1] the analogous gaps at $r = 22, 24, 26$ are filled by Construction QM_3^2 and $r = 28$ by QM_5^2, with $r = 42, 44$ following from the $r = 28$ code. The ingredient that QM_5^2 requires is present here, by the last assertion of Theorem 3, but those constructions are not carried out in this note and no claim is made for those r.

5 What is not claimed

The length 50 is not shown to be optimal: the sphere-covering bound gives only $\ell_2(10, 2) \geq 45$, since $1 + n + \binom{n}{2} \geq 1024$ fails at $n = 44$. Proposition 2(i) shows only that this particular 50-set has no redundant column. An annealing run at $n = 49$ terminated with 7 uncovered syndromes; that is an incomplete search and not a lower bound.

Theorems 1, 3 and 5 are established by complete enumeration. Corollary 6 is not: beyond $m = 4, 5$ it inherits the bound $p(\boldsymbol{H}_C) \leq 2^{m+1} + 1$ of [1] (4.4) rather than exhibiting the partitions, exactly as the corresponding argument in [1] does. Computing those partitions explicitly would strengthen the corollary.

Table 5.1 of [1] is stated to be best "as far as the authors know". We have not been able to search the ACCT proceedings exhaustively and make no priority claim for $\ell_2(10, 2) \le 50$.

Verification

The matrix of Appendix A, the partition of Theorem 3, and the two propagated matrices of Theorem 5 are accompanied by two independent verifiers, in Python and in Rust, which read the matrices from plain text and re-derive every number in this note by exhaustive enumeration: no stored certificate is consulted and no sampling is used. The two use opposite algorithms — one enumerates pairs and marks the syndromes they hit, the other sweeps the syndromes and searches for a partner column — and their outputs are compared automatically. The complete run takes about a second and is deterministic.

The repository is `https://github.com/wustep/maths`. To re-run:

```
git clone https://github.com/wustep/maths.git
cd maths/problems/covering/result
./run_all.sh
```

The matrix was located by targeted simulated annealing seeded from $\boldsymbol{H}_{KR}$, run by an automated agent. The search procedure is not part of the argument: Theorem 1 is a finite decidable assertion about a fixed matrix.

A The matrix $\boldsymbol{H}$

Rows 1 to 10 from top to bottom, columns left to right.

```
10010010011001010110111100110100100011110111111110
01010001001101000001001110111100011110001100001010
00110001011011000011111011101100111000111110101101
00010000010100010101010001110100110001001000101011
00001001001011000010110110101000010100110101101100
00000100011001010001110010011101001100010011100011
00000011000111001111110001111011011100001111100111
00000000111111000000001111111000111100000000011111
00000000000000111111111111111000000011111111111111
00000000000000000000000000000111111111111111111111
```

The same columns as unsigned integers, with bit i (least significant first) equal to row $i + 1$:

```
1 2 4 15 16 32 65 86 128 173 183 202 212 247 256 297 320 329 341
366 373 381 391 403 438 460 479 491 502 559 576 608 653 734 742
754 771 777 789 821 846 855 869 881 893 897 927 981 1003 1004
```

References

[1] A. A. Davydov, S. Marcugini, F. Pambianco, *New upper bounds for binary linear covering codes*, arXiv:2511.02542 [cs.IT], November 2025.

[2] M. Kaikkonen, C. Rosendahl, *New covering codes from an ADS-like construction*, IEEE Trans. Inform. Theory **49** (2003), no. 7, 1809–1812. `https://doi.org/10.1109/TIT.2003.813508`.

[3] G. Cohen, I. Honkala, S. Litsyn, A. Lobstein, *Covering Codes*, North-Holland Mathematical Library **54**, Elsevier, Amsterdam, 1997.

[4] B. Green, *100 open problems*, Problem 40. Available at `https://people.maths.ox.ac.uk/greenbj/papers/open-problems.pdf`.